

torch-spyre Ecosystem Health Report

May 21, 2026 | Pod: rganti-spyre-main | PyTorch 2.11.0

3/3 BUILD PASS	3/4 CI PASSING	20 PRS MERGED (7D)	30 OPEN BUGS
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Build Status

Component	Commit	Status	Notes
deeptools	b7f6fdbfa7 — Update dxp.cpp	PASS	cmake + make -j16 install
flex	bbea5e3 — add MemoryType to FlexAllocator	PASS	cmake + make -j16 install
torch-spyre	418cd9a — Finding reduction dimension in matmul/exx2/layernorm/topk	PASS	Required USE_SPYRE_CCL=0 (spyre_comms not installed)

CI Status

Workflow	Last Run	Status	Details
Spyre Hardware Tests	May 21 12:53 UTC	HEALTHY	In progress / awaiting approval
Linters	May 21 12:53 UTC	PASS	Consistently passing
Upstream PyTorch Tests	May 21 12:53 UTC	PASS	Consistently passing
Nightly Tests	May 21 03:15 UTC	FAIL	Persistently failing (3+ days), 2 tests only

Nightly Failure Triage

Failure 1: test_activation_fn_mish_fp16_lx_planning_reduction Suite: tests/inductor/test_inductor_ops_lx_planning.py — LxPlanningTwoOpReductionTest

Error: AssertionError — numerical tolerance exceeded (1/128 elements, abs diff 0.18 vs allowed 0.01)

Assessment: **FLAKY** — borderline fp16 precision issue with tight tolerance

Fix: Increase atol from 0.01 to 0.02, or seed the random input

Failure 2: test_instantiate_device_type_tests_mro

Suite: tests/test_spyre.py — TestSpyre

Error: AssertionError — expects class name "_TestMROCheckPRIVATEUSE1" but gets "_TestMROCheckSPYRE"

Assessment: **REGRESSION** — PyTorch 2.11 changed device naming from PRIVATEUSE1 to SPYRE

Fix: PR #2161 (branch fix-mro-test-pt211) is ready to merge

Context: Down from 648 failures on May 17 (test_modules_custom.py float64 issue fixed by commit 53dc945 on May 19). Only 2 failures remain — nightly is nearly green.

Version Alignment

Check	Expected	Actual	Status
PyTorch	~=2.11.0 (pyproject.toml)	2.11.0+cpu	MATCH

Repo Drift (Last 7 Days)

Repo	Commits	API-Touching	Risk
deeptools	91	10 (HCC, DCC, Dialects, DXP)	HIGH
flex	13	9 (C++20 upgrade, MemoryType, AllocationDirective)	HIGH
torch-spyre	56	—	—

Key upstream changes: flex C++20 upgrade (#1050), MemoryType in FlexAllocator (#1008), AllocationDirective required (#1032, matched by torch-spyre #2034). Deeptools: HCC dl16→fp32 legalization, DXP allocation node changes, Agen dialect reorganization.

Blocking PRs

PR	Title	Why Blocking	Review Status
#2190	Add memory type for allocate function	Allocator API change — needs deeptools coordination	REVIEW REQUIRED
#2135	Add senprog handling	New runtime integration point	REVIEW REQUIRED
#1943	Dualkineto integration	Profiler infrastructure, 16 days stalled	CHANGES REQUESTED

Cross-Repo API Mismatches

spyre_comms dependency: torch-spyre's CCL backend (spyre_ccl.cpp) requires `spyre_comms.hpp` which is not installed on the pod. Build succeeds with `USE_SPYRE_CCL=0`. Not a regression — existing gap in the single-card dev environment.

AllocationDirective: flex #1032 made this required; torch-spyre #2034 (merged May 19) already adapted. No mismatch.

Project Velocity

20 PRs merged (7d) **20** in review **23** stalled (>7d)

3 approved, ready to merge

Merge velocity: ~2.9 PRs/day. 3 PRs approved and ready to land (#2210, #2189, #2128). 8 dependabot PRs stalled 51 days (CI action bumps). Focus areas this week: matmul/attention fixes, FP32 enablement, test framework improvements.

Open Bugs

Metric	Count	Details
Total open	30	Majority: missing ops + numerical accuracy (LLM onboarding)
New (24h)	2	#2195 MHA attention compile failure, #2177 Mistral-Small scalar constant equality
Filed (7d)	5	Filing rate ~0.7/day
Closed (7d)	3	#1720 torch.neg, #1719 torch.cat, #1690 torch.getitem — all closed today
Net trend	+2	Backlog growing slowly; fix velocity accelerating

Smoke Test

Test	Status	Details
Import	PASS	torch.spyre.is_available() = True, device_count = 1
Eager execution	PASS	4x4 fp16 tensor add on Spyre
Compiled execution	PASS	128x128 fp16 torch.compile() add on Spyre

Action Items

- 1 Merge PR #2161** — fixes MRO test for PT 2.11 device naming. Will eliminate 1 of 2 nightly failures immediately.
- 2 Fix mish test tolerance** — increase atol from 0.01 to 0.02 in test_inductor_ops.py for the mish activation test. Will make nightly fully green.
- 3 Add USE_SPYRE_CCL=0 to activate.sh** — prevents CCL build failures on single-card pods without spyre_comms installed.
- 4 Land approved PRs** — #2210, #2189, #2128 are approved and waiting to merge.
- 5 Review #2190 (MemoryType allocator)** — cross-repo API change that needs coordination with deeptools/flex.
- 6 Clear dependabot backlog** — 8 CI action bump PRs stalled 51 days. Batch-merge to reduce noise.
- 7 Triage new bugs** — #2195 (MHA compile failure) and #2177 (Mistral scalar equality) filed in last 24h.